Homework

DM11 - A face and its altitude - 50 min.

In an orthonormal coordinate system with origin O, consider the points $A(2;0;0)$, $B(0;2;0)$ and $C(0;0;2)$.

1. Prove that the triangle ABC is equilateral and calculate its area.

2. Show that the vector $\vec{n} = (1;1;1)$ is normal to the plane (ABC).

3. Let H be the orthogonal projection of O onto (ABC). Justify that its coordinates are of the form $(t;t;t)$. Determine t using $\overrightarrow{AH}\cdot\vec{n}=0$.

4. Calculate OH, then the volume of the tetrahedron $OABC$ by taking ABC as base. Check this volume by taking OAB as base.